

Consolidated theorem manifesto: the universal spectral lower bound and its corollaries

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Setup (one paragraph)

Let $G = (V, E)$ be a finite undirected graph with $|V| = N$, symmetric-normalised Laplacian $L = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$. Let $\mathbf{y} \in \mathbb{R}^N$ be a centred bounded target signal, $\|\mathbf{y}\|_\infty \leq M$, $\mathbf{1}^\top \mathbf{y} = 0$, $\text{Var}(\mathbf{y}) = N^{-1}\|\mathbf{y}\|^2$. Let $\mathcal{A}$ be a learner with closed hypothesis class $\mathcal{H}_d \subseteq \mathbb{R}^N$ truncated to $[-M, M]$, ERM, with population-risk minimiser $f_{\mathcal{H}_d}^* := \arg \min_{f \in \mathcal{H}_d} L_V(f)$ and *learner-specific spectral R^2* $R_{\text{spec}}^2(\mathcal{H}_d, \mathbf{y}) := 1 - L_V(f_{\mathcal{H}_d}^*)/\text{Var}(\mathbf{y})$. Evaluation protocol Π_{LSO} is uniform-without-replacement sampling of a training set $T \subset V$ at $|T| = n$, $u := N - n$, $\rho(\Pi) := N/u$.

The results, in one table

Result	Statement	Status	Loc.
Theorem 1 (Universal LB)	$\mathbb{E}_T[L_{T^c}(\hat{f}_{\mathcal{A}}(T))] \geq (1 - R_{\text{spec}}^2(\mathcal{H}_d, \mathbf{y})) \cdot \text{Var}(\mathbf{y})$, exact in expectation	proved	paper §5; notes/01
Theorem 2 (Sharp transductive UB)	$L_{T^c}(\hat{f}_{\mathcal{A}}(T)) \leq$ $(1 - R_{\text{spec}}^2)\text{Var}(\mathbf{y}) +$ $2\mathcal{R}_n^{\text{trans}}(\mathcal{H}_d) +$ $5.05M^2\sqrt{\frac{(n+u)\log(2/\eta)}{nu}}$, with $\mathcal{R}_n^{\text{trans}}$ the transductive Rademacher of El-Yaniv and Pechyony (2009)	proved	paper §6; notes/02, 03, 04
Theorem 3 (Berry-Esseen minimax)	$\mathbb{P}_T[L_{T^c}(\hat{f}) - (1 - R_{\text{spec}}^2)\text{Var}(\mathbf{y}) \geq cM^2/\sqrt{u}] \geq 1 - \eta$ for some signal $\mathbf{y} \in \mathcal{H}_d^\perp$; uniform in $\hat{f}$, even oracle	proved	paper §7; notes/02
Corollary 1 (Cramér-Rao analog)	Combining Thm 1 + Thm 2 + Thm 3: minimax slack rate is $\Theta(M^2/\sqrt{u} + \mathcal{R}_n^{\text{trans}}(\mathcal{H}_d))$, matching to a factor ≤ 10	direct	paper §4
Corollary C1 (Encoding gap)	$\mathbb{E}_T[L_{T^c}(\hat{f}_{\text{cat}}) - L_{T^c}(\hat{f}_{\text{comp}})] \geq$ $R_{\text{spec}}^2(\mathcal{H}_{\text{comp}}, \mathbf{y}) \cdot \text{Var}(\mathbf{y})$, recovering the MLST paper exactly	proved	paper §8.1
Theorem 6 / C2 (Negative transfer)	Under spectral disjointness of $(\mathbf{y}_{\text{src}}, \mathbf{y}_{\text{tgt}})$ on $\mathcal{H}_d$: $\mathbb{E}[\text{NT}(\mathcal{A})] \geq$ $(\ \mathbf{P}_{\mathcal{H}_d}\tilde{\mathbf{y}}_{\text{src}}\ ^2 + \ \mathbf{P}_{\mathcal{H}_d}\tilde{\mathbf{y}}_{\text{tgt}}\ ^2)/n_t$. Transfer is unavoidably harmful.	proved	paper §8.2
Corollary 2 (NT vacuity)	Under spectral alignment of $(\mathbf{y}_{\text{src}}, \mathbf{y}_{\text{tgt}})$ on $\mathcal{H}_d$, $\mathbb{E}[\text{NT}(\mathcal{A})] = 0$	direct	paper §8.2
Theorem 7 / C3 (Active learning)	Importance-weighted ERM under leverage-score sampling $\pi^*(v) \propto \ \mathbf{B}(v, \cdot)\ ^2$ achieves slack $O(M^2\sqrt{dN\log(1/\eta)/n})$, i.e. rate $1/n$ (parametric) instead of passive $1/\sqrt{n}$	proved	paper §8.3
Withdrawn (Conj. 1, notes/02 §3)	Non-realisable minimax rate has factor σ_Π^{-1}	withdrawn	notes/03 §4–5
Withdrawn (“Thm 2”, notes/01)	Saturation slack scales as $\rho(\Pi)/\sqrt{n\sigma_\Pi}$	withdrawn	notes/02 §1 (erratum)

Resolved open questions (post notes/01–04)

1. **Saturation of the lower bound.** Resolved by Thm 2. Pesenson-ridge is not needed; the population-risk minimiser $f_{\mathcal{H}_d}^*$ serves as a clean witness (notes/03).
2. **Non-realisable regime.** Vacuous (notes/03). No separate minimax theory required.
3. **Sharp $\rho(\Pi)$ constant.** Resolved: ρ was loose. Transductive Rademacher (El-Yaniv and Pechyony, 2009) eliminates it (notes/04). Sharp constant within a factor of ≤ 10 ; multi-point Fano (sketched, notes/04 §4) would tighten to ≤ 4 .

Remaining (non-foundational) open questions

1. Exact Fano multi-point constants in Theorem 3 (cosmetic refinement; deferred).
2. Importance-weighted transduction identity for general non-uniform sampling distributions (mechanical extension required for C3, partially done in paper §8.3).
3. Empirical anchor for C2 (negative transfer) on a QM9 $\rightarrow$ MatBench task pair (companion repo `negative-transfer-cor` in preparation).
4. Empirical anchor for C3 (active learning) showing the $\sqrt{n/d}$ improvement of leverage-score over uniform sampling (companion repo `active-learning-corollary` in preparation).

Where to start reading

- **Newcomers:** start with the main paper’s `paper.tex` §1 (Introduction) and §4 (Main results) for the headline statements, then §8 (Corollaries) for the empirical instantiation. The Cramér-Rao analogy is established in §4 (Corollary 1).
- **Collaborators reviewing the proofs:** paper §5–§8 contain the full proofs (transductive Rademacher, Berry-Esseen, NT, AL). For the historical development and superseded statements, see notes/01–04 in order; notes/05 (this document) is the consolidated summary.
- **Reviewers:** `cover_letter.md` positions the contribution. Falsifiability section (paper §9.2) lists three pre-registered statistical tests; F1 passes with exact $p = 4.96 \times 10^{-5}$ on the 8-task MatBench panel. The SI (`paper_si.tex`, 5 appendices) contains technical details deferred from the main paper.

References

Ran El-Yaniv and Dmitry Pechyony. Transductive Rademacher complexity and its applications. *Journal of Machine Learning Research*, 10:193–234, 2009. Conference version: COLT 2007.